Huawei Access Network QoS Congestion Event Standard Operating Procedure (SOP)

Node: BTM-ACC-23 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_1086

Description

This Standard Operating Procedure (SOP) addresses a Critical QoS Congestion Event on a Huawei Access Network device, specifically Node ID BTM-ACC-23, at the BTM Layout Aggregation site. The failure is caused by Microbursts, interface tail drops, and QoS queue starvation, resulting in Alarm Code ALARM_1086. The underlying protocol mechanics involve the Huawei Access Network device's QoS policy, which is implemented using the Multi-Field Classification (MFC) and Policer functions. The MFC function classifies traffic based on multiple packet fields, including 5-tuple information, VLAN ID, and COS, while the Policer function enforces rate limits and burst sizes for each traffic class. When the QoS policy is misconfigured or overwhelmed by traffic bursts, it can lead to QoS queue starvation, causing network congestion and triggering the ALARM_1086.

Pre-requisite Checks

- Task: Verify the network topology and identify redundant links, Description: Ensure that redundant links are available and functioning correctly to minimize network downtime during the troubleshooting process.

- Task: Dispatch remote hands to the BTM Layout Aggregation site, Description: Send a technician to the site to perform physical checks and provide hands-on assistance during the troubleshooting process.

- Task: Check the device's interface status and QoS policy configuration, Description: Verify that all interfaces are up and running, and that the QoS policy is correctly configured to handle the expected traffic load.

Potential Root Causes

Cause	Probability
QoS policy misconfiguration	High
Traffic burst overload	Medium
Interface hardware failure	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policer, Expected Output: Policer ID: 1 Policer Name: policer-1 Policer Type: rate-based Rate: 1000 bps Burst: 1000 bytes, Description: Display the policer configuration to verify the rate limits and burst sizes enforced by the QoS policy.
```

```
Command: display qos mfc, Expected Output: MFC ID: 1 MFC Name: mfc-1
Match Criteria: 5-tuple, VLAN ID, COS Action: policer-1, Description:
Display the MFC configuration to verify the classification criteria
and corresponding policer assignments.

Command: display interface, Expected Output: Interface:
GigabitEthernet0/0/1 Status: up MTU: 1500 bytes, Description: Display
the interface status to verify that it is up and running.
```

Resolution Steps

Step 1: Step: Review and adjust the QoS policy configuration, Command: configure qos policy policer rate burst , Description: Review the QoS policy configuration to ensure that it is correctly set up to handle the expected traffic load. Adjust the policer configuration as needed to prevent QoS queue starvation.

Step 2: Step: Verify that the QoS policy is correctly applied, Command: display qos policer, Description: Verify that the QoS policy is correctly applied and that the policer configuration matches the expected settings.

Step 3: Step: Check for any software or hardware issues, Command: display diagnose, Description: Check for any software or hardware issues that may be contributing to the QoS congestion event.

Post-Check Verification

```
Command: display qos policer, Expected Output: Policer ID: 1 Policer
Name: policer-1 Policer Type: rate-based Rate: 1000 bps Burst: 1000
bytes, Description: Verify that the policer configuration is correctly
set up to prevent QoS queue starvation.

Command: display qos mfc, Expected Output: MFC ID: 1 MFC Name: mfc-1
Match Criteria: 5-tuple, VLAN ID, COS Action: policer-1, Description:
Verify that the MFC configuration is correctly set up to classify
traffic and assign policers.

Command: display interface, Expected Output: Interface:
GigabitEthernet0/0/1 Status: up MTU: 1500 bytes, Description: Verify
that the interface is up and running.
```

Roll-back Procedures

- Step: Restore the previous QoS policy configuration, Command: configure qos policy policer rate burst rollback, Description: Restore the previous QoS policy configuration if the fix fails.

- Step: Verify that the QoS policy configuration is restored to the previous state, Command: display qos policer, Description: Verify that the QoS policy configuration is restored to the previous state.

- Step: Check for any software or hardware issues, Command: display diagnose, Description: Check for any software or hardware issues that may be contributing to the QoS congestion event.